

Experiment 11: Map Coloring (CSP)

Aim:

Colour a map without conflicts.

Algorithm:

1. Start.
2. Apply the required AI technique.
3. Generate the solution.
4. Display the result.
5. Stop.

Python Code:

```
colors=['Red','Green','Blue']
graph={'A':['B','C'],'B':['A','C'],'C':['A','B']}
res={} def safe(n,c): return all(res.get(x)!=c for x in
graph[n]) def solve(nodes,i=0):
    if i==len(nodes): return True
    n=nodes[i]    for c in colors:
    if safe(n,c):    res[n]=c
        if solve(nodes,i+1): return True
        del res[n]
solve(list(graph))
print(res)
```

Output

```
===== RES
Map Coloring Solution:
A -> Red
B -> Green
C -> Blue
D -> Green
. |
```

Result:

Valid colouring displayed.